

2023



Upping Performance by 40% and Optimizing Cost Using Amazon OpenSearch Service with Dream11

Learn how Dream11 serves over 200 million online gamers by reimagining its analytics workloads using Amazon OpenSearch Service.

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Nearly 50% reduction

in compute costs

Reduced hours

of manual scaling

40% improvement

in cluster performance

30% reduction

in node use

Enhanced

internal



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Overview

As an extremely popular global fantasy sports platform, [Dream11](#) provides players with near-real-time updates and recommendations based on the statistics of real athletes. The management of its Elasticsearch is costly and complex, and engineers spent hours scaling manually to accommodate fluctuating



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Elasticsearch announced the sunsetting of its open-source distribution, Dream11 researched other options to power the search and analytics engine that is critical to its business.

Dream11 chose to adopt a solution managed by Amazon Web Services (AWS) so that it could run and scale clusters without having to worry about managing, monitoring, and maintaining the infrastructure. It migrated to [Amazon OpenSearch Service](#), an open-source, distributed search and analytics suite derived from Elasticsearch. Working alongside AWS experts to fine-tune its compute infrastructure, Dream11 significantly optimized costs, boosted performance by 40 percent, and built an enhanced experience for its 200 million users.



Opportunity | Using Amazon OpenSearch Service to Manage Search and Analytics for Dream11

Based in Mumbai, India, Dream11 brings Indian sports fans closer to the sports they love by providing world-class fan engagement and user experience through fantasy sports. The Dream11 app lets users create their own teams and play fantasy cricket, soccer, kabaddi, basketball, hockey, volleyball, handball, rugby, futsal, American football, and baseball. To select players for their fantasy teams, users need to have unfettered access to up-to-date statistics, especially before and after real-world matches. As a result, Dream11 experiences heavy spikes in workloads, and these would strain CPU utilization unevenly as the system took in terabytes of data. Some CPUs reached 95 percent utilization, while others hovered around 6 percent.

Dream11 had been running its search and analytics capabilities on Elasticsearch. It managed its own instances of [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for any workload. Manual scaling of clusters contributed to high operational overhead, and the cost of adding instances to accommodate frequent peaks in use resulted in high compute costs.



In 2021, Elasticsearch announced that it was phasing out its open-source distribution. Dream11 could no longer update its clusters, and it started to experience lagging speed and other performance issues. The company faced two options: migrate to a commercial license or switch to a forked version of Elasticsearch. Dream11 chose to implement OpenSearch, which remained an open-source and community-driven project. “We did our own research and found that Amazon OpenSearch Service was quite flexible, and its pricing was similar to the cost of managing our own OpenSearch machines on AWS,” says Mehul Batra, software engineer at Dream11.

Solution | Improving Cluster Performance by 40 Percent While Significantly Optimizing Compute Costs

For the migration, Dream11 received support and guidance from AWS experts who helped optimize cluster configuration, performance, and security based on operational metrics and best practices. For example, Dream11 implemented Amazon EC2 instances powered by [AWS Graviton Processors](#), designed by AWS to deliver the best price performance for cloud workloads running in Amazon EC2. Using AWS Graviton processor-based instances, Dream11 improved cluster performance by 40 percent while using 30 percent fewer nodes. The company optimized compute costs significantly.

Dream11 engineers no longer need to add up to 100 instances to handle peak demand. Instead, they scale vertically, using the more powerful processors to distribute CPU utilization more uniformly among fewer machines. “We realized that we were getting better performance using the AWS Graviton processor-based machines specific to OpenSearch, which are built for the search engines,” Batra says. “Our CPUs are more stable.”

Even at a peak of 40,000 concurrent queries, players no longer experience time-outs or long delays. Previously, Dream11 analysts, data scientists, or product teams sometimes had to wait up to 10 seconds for data to refresh. Now, every query returns results within 150 milliseconds of latency, which has made it about 98 percent quicker for internal users to see updated statistics.

Figure 1: Comparison of Dream11’s use of Elasticsearch and Amazon OpenSearch Service

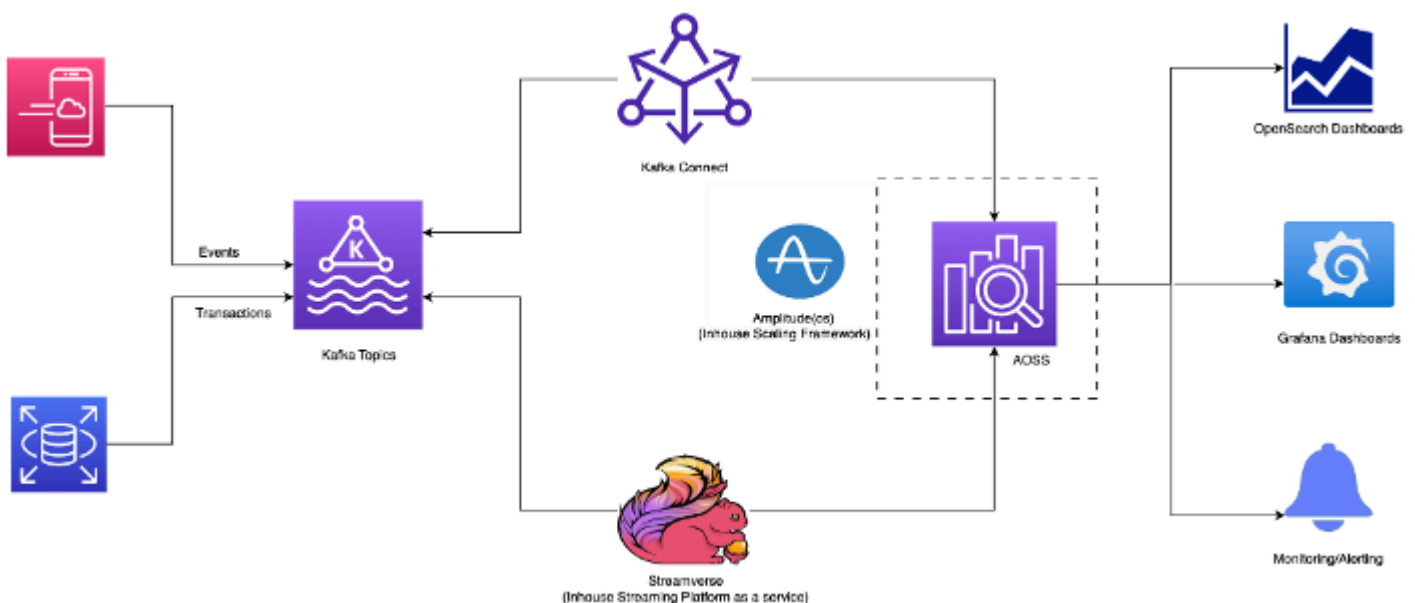


Benefits	Elasticsearch	Amazon OpenSearch Service
Peak CPU (Average)	80%	45%
Peak Latency (Average)	1000 ms	150 ms
Shard Size (Concurrent Reads and Writes)	30	10
Replica	1	2
Scaling	Manual	Automated
Tenancy Support	No	Yes
Governance	No	Read and Write
Lifecycle Policy	Manual	Automated
Rollover Policy	Manual	Automated
Data Freshness	Manual	Automated
Alias Policy	Manual	Automated

Using the managed scaling built into Amazon OpenSearch Service, Dream11 significantly cut its operational overhead. Engineers no longer spend many hours manually scaling up at peak and scaling down after matches. With their time savings, Dream11 engineers focus on improving search and analytics capabilities for internal users by building customized visualizations and reports to unlock valuable insights. Additionally, engineers use predefined features of Amazon OpenSearch Service with reusable drag-and-drop components to manage the cluster's lifecycle policy, rollover policy, alias creation, and alerting, which used to require custom scripts.

It also built fine-grained access control into dashboards while keeping the functionality simple for internal users as they track key performance indicators. Dream11 established a tenant-based architecture so that different persona users—data analysts or scientists, for example—access only the dashboards for which they have permission. Users maintain the single sign-on that they had with the previous system, simply entering their emails to log on or query data without the need for a username and password. “That was something that was missing in our old cluster,” Batra says. “We really wanted to use it in OpenSearch so that users wouldn’t interfere with each other’s privacy.”

Figure 2: Dream11 architecture diagram



Outcome | Enhancing the Internal User Experience Using OpenSearch Functionality

To further enhance the internal user experience and generate analytics to drive business outcomes, Dream11 plans to implement additional features of OpenSearch. For example, OpenSearch’s Machine Learning Commons library provides algorithms to train models and predict trends in data. Dream11 will also use machine learning for anomaly detection, a feature of OpenSearch that automatically detects anomalies as the system ingests data and then uses the alerting functionality to send notifications in near real time. “We chose Amazon OpenSearch Service for the perks of OpenSearch with the robustness of automatic updates, managed scaling, and AWS community support,” Batra says. “Now, our data analysts and scientists happily live with their data and dashboards.”

About Dream11

The flagship brand of leading Indian sports technology company Dream Sports, Dream11 is a fantasy sports platform with more than 200 million users playing 11 sports on the app.

AWS Services Used

Amazon OpenSearch Service

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon Elastic Compute Cloud (Amazon EC2)

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

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AWS Graviton Processors



AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

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